# E1 213 Pattern Recognition and Neural Network
# Assignment 2

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## Phase 1: The MLP Baseline & Gradient Flow (Air Quality)

### 1.1 Flattened Temporal MLP (Regression)

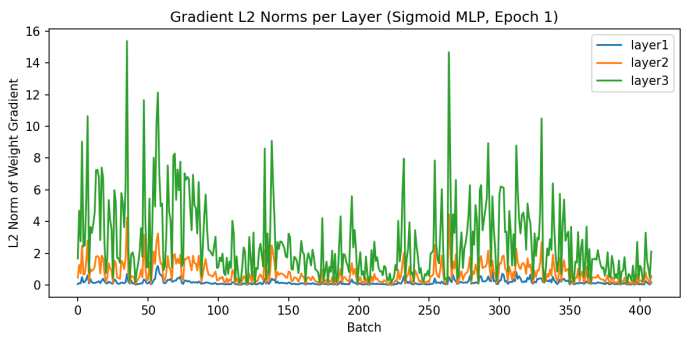
**Ans:** A 3-layer MLP is trained on 72-hour flattened feature vectors to predict $t$+24 PM2.5. The data is split chronologically (70/15/15) with train-split normalization to prevent leakage. Test MSE (normalized) = **0.8568**.

### 1.2 The Vanishing Gradient Proof

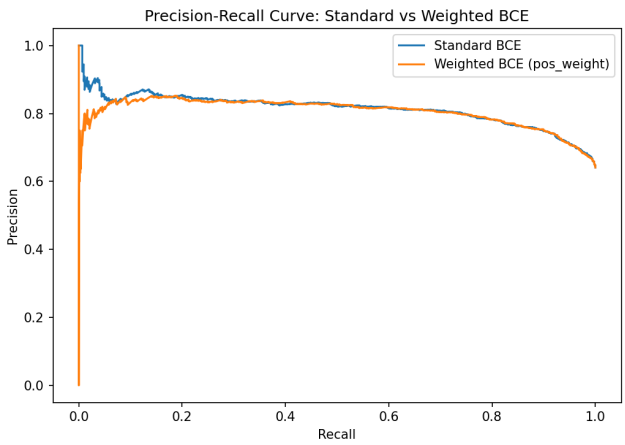
**Ans:** The sigmoid activation $\sigma(x)$ has derivative $\sigma'(x) = \sigma(x)(1-\sigma(x)) \leq 0.25$. Backpropagation multiplies these derivatives across layers, so the gradient of layer 1 scales as $(0.25)^3 \approx 0.016$ relative to the output layer. Fig. 1a confirms this: the $L_2$ gradient norm of the first layer is orders of magnitude smaller than the third.

### 1.3 Imbalanced Threshold Classification

**Ans:** PM2.5 > 200 is labelled hazardous (class 1). Standard BCE gives recall of 61% for the minority class, while Weighted BCE (pos_weight $\approx$ 2.86) improves minority recall to 78% at the cost of some precision for a threshold of 0.5. Interesting thing is that the PR curve looks almost similar. That is because the training and testing has very different demographic. The testing has much more PM2.5 > 200 cases than the train one.




(a) $L_2$ gradient norm per layer (Sigmoid MLP, Epoch 1)




(b) Precision-Recall: Standard vs. Weighted BCE

## Phase 2: CNNs & Spatial Hierarchies (PlantVillage)

### 2.4 CNN from Scratch & Receptive Fields

**Ans:** The CNN has three Conv-BN-MaxPool blocks (8, 16, 32 filters) followed by three linear layers. Tensor shapes for a $128 \times 128$ input: $(1,3,128,128) \to (1,8,128,128) \to (1,8,64,64) \to (1,16,64,64) \to (1,16,32,32) \to (1,32,32,32) \to (1,32,16,16) \to (1,8192) \to (1,256) \to (1,38)$. The theoretical receptive field of a final feature-map activation is **22 pixels**, computed as $\mathrm{RF}_n = \mathrm{RF}_{n-1} + (k-1)\prod_{i<n} s_i$ iterated over the three conv+pool stages.

### 2.5 CNN Regression Head

**Ans:** The classification head is replaced with a single linear output and MSE loss. The severity target is synthesized as the bounding-box ratio of necrotic pixels to total leaf area. Test MAE = **0.2392**.

### 2.6 Translation Invariance

**Ans:** Baseline accuracy on 100 test images: **86.00%**. After a 5-pixel shift: **77.00%** — a drop of 9%. Max-pooling provides only local invariance within each $2 \times 2$ window. A 5-pixel shift moves features across pooling-cell boundaries, changing which neuron "wins," so the downstream representation changes and exact translation invariance is not achieved.

## Phase 3: RNNs & Sequential Memory (Air Quality)

### 3.7 Vanilla RNN from Scratch

**Ans:** A single-layer RNN is built using `nn.Linear` layers and a `tanh` loop: $h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t + b)$. Trained with Adam (lr $= 10^{-3}$, hidden=64) for 100 epochs with early stopping. Test MSE (normalized) = **0.9198**.

### 3.8 BPTT Decay

**Ans:** Gradient magnitudes extracted via `retain_grad()` on CPU: $\|\partial L/\partial h_{100}\| = 1.79 \times 10^{-1}$, $\|\partial L/\partial h_{50}\| \approx 0$, $\|\partial L/\partial h_0\| \approx 0$. The gradient scales as $\|W_{hh}^\top\|^{T-t}$; since the spectral radius $< 1$ for a stable network, repeated multiplication drives gradients to zero exponentially. Fig. 2a shows this alongside the LSTM for direct comparison.

### 3.9 Exploding Gradients

**Ans:** With SGD at lr = 10, the loss reaches NaN after 1 epoch. The maximum singular value of $W_{hh}$ just before the crash is $\boldsymbol{\sigma_{\max}} = \mathbf{1.1057}$. For $\sigma_{\max} > 1$, the gradient grows as $\sigma_{\max}^T$: over $T = 72$ steps, $1.1057^{72} \approx 1.8 \times 10^3$, mathematically guaranteeing unbounded gradient growth without clipping.

## Phase 4: LSTMs, GRUs, & Context (Air Quality)

### 4.10 LSTM Gradient Rescue

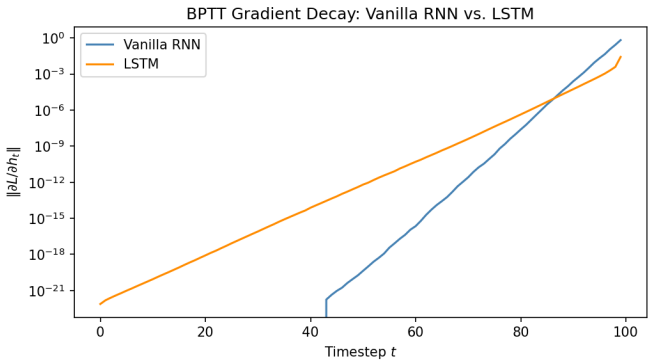
**Ans:** LSTM gradients: $\|\partial L/\partial h_{100}\| = 1.32 \times 10^{-1}$, $\|\partial L/\partial h_{50}\| = 1.17 \times 10^{-12}$, $\|\partial L/\partial h_0\| = 1.59 \times 10^{-22}$. Compared to the vanilla RNN (which was already zero at $t = 50$). The forget gate $f_t \in (0,1)$ controls the cell gradient: $\partial c_t/\partial c_{t-1} = f_t$, which can remain close to 1, allowing gradients to flow further back than in a vanilla RNN.

### 4.11 GRU vs. LSTM Efficiency

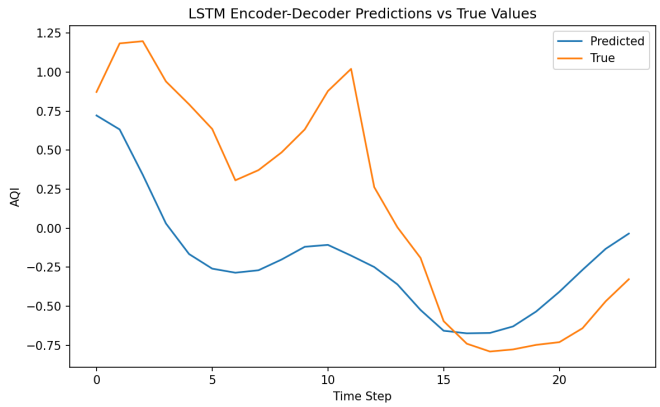
**Ans:** LSTM trainable parameters: **67,201**; GRU: **50,433** ($\approx 25\%$ fewer, as GRU uses 2 gates vs. LSTM's 3). Best validation MSE — LSTM: 0.2530, GRU: 0.2480. Training time — LSTM: 45.88 s for 11 epochs, GRU: 99.19 s for 10 epochs. Test MSE — LSTM: 0.8795, GRU: 0.9690. GRU achieves marginally better validation loss with fewer parameters.

### 4.12 Sequence-to-Sequence Forecasting

**Ans:** An Encoder-Decoder LSTM encodes 72 hours into a context vector; the decoder autoregressively predicts the full 24-step trajectory. Test MSE (normalized) = **0.6591**. Fig. 2b shows the ground-truth vs. predicted 24-hour PM2.5 trajectory on a test sample.




(a) BPTT gradient decay: Vanilla RNN vs. LSTM (log scale)




(b) Encoder-Decoder LSTM: ground truth vs. 24-step prediction

## Phase 5: Transformers & Attention

### 5.13 Scaled Dot-Product Attention

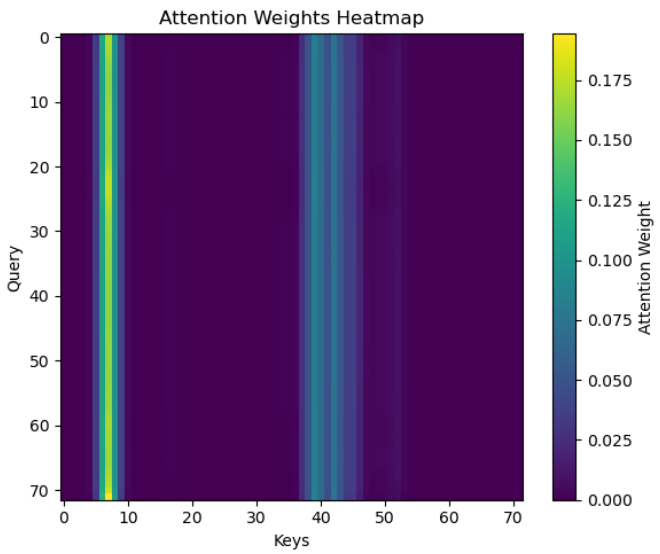
**Ans:** Test MSE = **1.1395**. Fig. 3a shows the $72 \times 72$ attention weight heatmap; without positional encoding the weights collapse to a near-uniform column, indicating the model cannot distinguish temporal position.

### 5.14 The Necessity of Positional Encoding

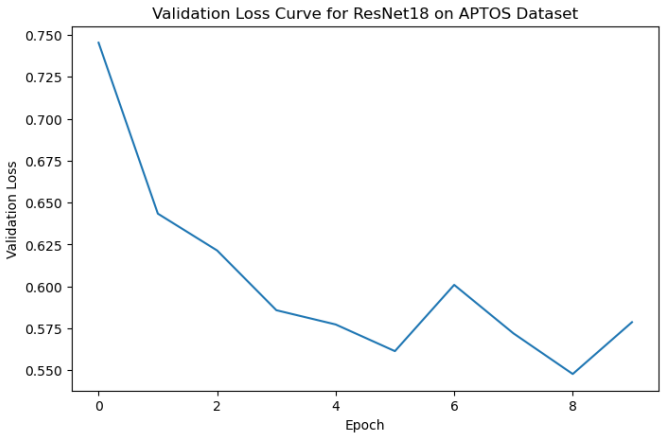
**Ans:** Without PE: best val MSE = 0.2344, test MSE = **1.1496**. With sinusoidal PE: best val MSE = 0.2295, test MSE = **0.8872**. Self-attention is permutation-invariant — $\text{Attention}(\pi Q, \pi K, \pi V) = \pi\,\text{Attention}(Q, K, V)$ — so any permutation of the input yields the same output. Without PE, the model fundamentally cannot represent temporal order; adding $\text{PE}(t, 2i) = \sin(t/10000^{2i/d})$ breaks this symmetry and injects positional information.

### 5.15 Vision Transformer (ViT)

**Ans:** Each $128 \times 128$ image is split into 64 non-overlapping $16 \times 16$ patches, each flattened to a 768-dim vector, giving a $(64, 768)$ sequence. The same single-head Transformer classifies disease across 38 classes. CNN accuracy: **86.55%** with 2,122,470 parameters (efficiency $4.1 \times 10^{-7}$). Attention accuracy: **42.43%** with 64,166 parameters (efficiency $6.6 \times 10^{-6}$). The ViT is $\approx 16\times$ more parameter-efficient.




(a) $72 \times 72$ attention weight heatmap (single head, no PE)




(b) Validation loss curve: ResNet18 fine-tuning (APTOS 2019)

# Phase 6: Advanced Vision & Imbalance (APTOS 2019)

### 6.16 Transfer Learning & Freezing

**Ans:** ResNet18 pretrained on ImageNet is loaded; all convolutional layers are frozen. The final FC layer is replaced with `nn.Linear(512, 5)`. Trainable parameters: **2,565**; total parameters: 11,179,077. Test accuracy: **77.45%**, Cohen's Kappa: **0.6404**. Fig. 3b shows the validation loss curve converging within 10 epochs.

### 6.17 Ordinal Imbalance & Focal Loss

**Ans:** The loss is implemented from scratch: $\mathcal{L}_{\text{focal}} = (1 - p_t)^\gamma \cdot \mathcal{L}_{\text{CE}}$ with $\gamma = 2$, where $p_t = e^{-\mathcal{L}_{\text{CE}}}$. This down-weights well-classified easy examples (large $p_t$) and forces training to focus on hard minority classes. Focal loss test accuracy: **78.00%**, Cohen's Kappa: **0.6496**, vs. cross-entropy accuracy: 77.45%, Kappa: 0.6404 — a consistent improvement on the ordinal metric. Standard cross-entropy accuracy is misleading because Class 0 dominates; a model predicting only Class 0 achieves high accuracy but zero clinical utility. Cohen's QWK penalizes predictions proportionally to their ordinal distance from the true label, making it the appropriate metric for this task.